Republic of Rwanda
Ministry of Health

Scope of Work for Quality Management Improvement Approach

1. Background and Rationale

Positive health outcomes depend on the effective functioning of the health system, including health information systems, financing, human resources, and the availability and management of medicines and other health commodities. Timely availability of quality-assured medicines at health facilities is essential to ensuring continuity and quality of patient care, and is directly influenced by the performance of the pharmaceutical supply chain.

Rwanda operates a decentralized pharmaceutical supply chain extending from the central level through Rwanda Medical Supply (RMS) Branches to service delivery points, including health centers and hospitals, and to community-level services. The system is designed to ensure that quality-assured medicines and health commodities reach patients at the right time, in the right quantity, in the right condition, and at an affordable cost.

Over the years, interventions implemented by the MoH together with supply chain stakeholders, have contributed to improvements in supply chain performance. However, sustaining these gains requires continuous performance monitoring, mentorship, supervision, and systematic quality improvement. In this context, the MoH, in collaboration with supply chain stakeholders, conducts the Quality Management Improvement Approach (QMIA) to identify performance gaps and understand their root causes to implement corrective actions and monitor improvement across key supply chain functions. For this year, the activity will be conducted jointly activity with the Ministry of Health (MoH), Rwanda Medical Supply (RMS), as well as Rwanda Social Security Board (RSSB).

In preparation for this year's assessment, Rwanda Medical Supply (RMS) and Rwanda Social Security Board (RSSB) were also planning to conduct separate prescription assessments to generate evidence on the availability and dispensing of prescribed medicines. To avoid duplication of efforts, optimize the use of available resources, and maximize the value of the assessments, the routine QMIA has been strategically aligned with these planned prescription assessments. Accordingly, the objectives and scope of the QMIA have been reviewed, where appropriate, to address the respective assessment objectives while maintaining a comprehensive approach to quality improvement and supply chain performance. This alignment will enable the participating institutions to undertake a coordinated assessment, generate complementary evidence, and develop stronger and more actionable recommendations for improving medicine availability and access.

Within this framework, medicine availability is commonly assessed as a stock measure, reflecting the proportion of tracer medicines physically available at a health facility at the time of assessment. While this indicator is important, it provides only a partial picture of access to medicines. It indicates

what is available on the shelf but does not necessarily show whether medicines prescribed to patients are actually dispensed.

An unfulfilled prescription represents more than a stock-out indicator; it is a direct service-delivery and patient-access event. When a prescribed medicine is unavailable, a patient may be required to purchase it from a private outlet, return to the facility at a later date, seek care elsewhere, or remain without the prescribed treatment. This may increase out-of-pocket expenditure, interrupt treatment, and ultimately affect health outcomes. However, because routine reporting does not systematically link prescription data with medicine availability and stock information across supply chain levels, the magnitude and value of unfulfilled prescriptions are not currently quantified at national level.

This activity will therefore complement the conventional measurement of stock availability by examining whether prescribed medicines are available and actually dispensed to patients, while also tracing the availability of the same medicines across the supply chain. This will provide a more comprehensive understanding of medicines access and help distinguish between different types of supply chain constraints.

The findings will provide evidence to guide quality improvement interventions, strengthen supply chain responsiveness, and ultimately improve access to medicines and continuity of care for patients.

2. Objectives

The assessment aims to quantify medicine availability and the volume, causes, and cost of unserved prescriptions across all levels of Rwanda's health system, and to attribute each gap to the point in the supply chain where it arises. Specifically, it will:

1. Measure availability of a defined tracer medicine list on the day of visit, by level of care, province, and district.
2. Determine the stock accuracy by comparing the physical stock versus stock recorded into digital tool used at each health facility.
3. Quantify stock-outs, current duration and frequency over the preceding three months.
4. Measure the proportion of prescriptions that go unserved and the quantity of medicine involved, across outpatient and inpatient services.
5. Determine whether each unserved medicine was available at RMS regional and central warehouses on the prescription date, separating distribution challenges from nationwide stock-outs.
6. Establish whether the facility requested the missing medicine and whether it was supplied fully, partially, or not at all.
7. Estimate the financial value of unserved prescriptions using official tariff prices.
8. Describe availability at community level through the community health worker attached to each selected facility.

3. Scope and Methodology

The assessment covers 267 health facilities and 237 community health workers across all 30 districts (504 visits in total) over 10 working days of fieldwork, spanning every level of care from second-

generation health posts to referral and teaching hospitals. It draws on facility records, RMS warehouse data, and RSSB claims data.

A multistage stratified and cluster sampling approach is used. The sample size of 237 facilities is derived using the Yamane formula (N = 583; 95% confidence; 5% margin of error), then allocated proportionally by level of care, province, and district, with facilities selected by computer-generated random sampling. A further 30 second-generation health posts, one per district, are added on top of the calculated sample to ensure every district is represented at the most peripheral tier, bringing the total to 267 facilities.

Where prescriptions are assessed, 30 to 50 per facility are drawn (scaled to level of care): 10 from the visit day and the remainder from the preceding 10 days, across outpatient and inpatient services. A defined medicine list, based on the country's top diagnoses, will cover both essential medicines and program commodities.

4. Data Collection Team and Logistics

The team combines central-level technical staff with locally stationed RMS branch managers, organized into 27 two-person teams, each with one vehicle. Each team is led by a central-level member and the RMS branch officer for the district being assessed will facilitate during the course within their respective districts. Central-level staff are drawn from MoH, RMS, Rwanda Biomedical Centre, RSSB, and CHAI. Each team covers 8 to 11 facilities plus attached community health workers over the 10 days, at roughly two visits a day.

5. Data Sources

Data are drawn from three sources depending on the variable: the health facility (physical counts, stock cards, dispensing logs, prescription registers, requisition records), RMS (regional and central warehouse stock, tariff prices, delivery records), and RSSB (claims and reimbursement data). Combining facility, RMS, and RSSB data allows each unserved prescription to be classified automatically as a distribution challenge or a nationwide stock-out, and valued at official tariff prices.

6. Timeline

The following table summarizes the proposed schedule for planned activities:

Table 1: Schedule for activities

Dates	Activity	Duration
21 August 2026	Pilot testing of tools at three Kigali facilities	1 working day
24–28 August 2026	Revision of data collection tools based on pilot testing findings	5 working days
9–11 September 2026	Training of data collectors	3 working days
14–25 September 2026	National field data collection (267 facilities, 30 districts)	10 working days
28 September- 9 October 2026	Data cleaning workshop	10 working days
12–30 October 2026	Data analysis and report writing	15 working days
6 November 2026	Dissemination workshop — presentation of findings	1 working day

7. Budget and Financing

The estimated cost of the assessment is **81,984,750 RWF** (approximately US$ **55,771.94**). This is a joint activity, and the budget is shared between the participating institutions (MoH, RMS and CHAI).

Through UNFPA support, the MoH will facilitate transportation for the data collection teams deployed from the central level. The estimated transportation cost is **42,639,750 RWF** based on the deployment of **27 vehicles for 10 days of fieldwork**. The remaining budget requirements will be covered by the other participating institutions, in accordance with the agreed cost-sharing arrangements.

Table 2: Summary budget

Budget item	Total (RWF)
Training of data collectors (Kigali, 3 days)	12,870,000
Transport — pilot (14 cars, 1 day)	3,570,000
Field transport for data collection teams (27 cars × 10 days)	42,639,750
Per diems — field data collection (central + RMS branch)	19,380,000
Data processing and analysis	625,000
Dissemination workshop	1,350,000
Grand total	**81,984,750**

Note: The other stakeholders involved in the activity will contribute to the budget and Ministry of Health will only cover the transportation of data collection team.

9. Central level participants

The central level participants in the field visits are the ones familiar with supply chain data quality assessment and shall come from various institutions is illustrated in the table below:

Table 3: Participants from Central

Participants positions	# of participants	Institution
Supply chain team	2	MoH/CPHSD
Supply Chain Quantification Specialist	2	RBC
Medical Supply Chain Senior Officer	1	RBC/HIV
Malaria Commodities management Specialist	1	RBC/Malaria Div.
Pharmacist Care and treatment	1	RBC/HIV
TB commodities staff	1	RBC/TB
Cancer Commodities Supply Chain Specialist	1	RBC/NCDs
Staff	2	RSSB
Staff	2	CHAI
Staff	11	RMS

The remaining participants will come from RMS branches

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